Atopic dermatitis

Atopic dermatitis (**AD**), also known as **atopic eczema**, is a long-term type of inflammation of the skin.[2] Atopic dermatitis is also often called simply **eczema**, but "eczema" is also used to refer to dermatitis, the larger group of skin conditions.[2][5] Atopic dermatitis results in itchy, red, swollen, and cracked skin.[2] Clear fluid may come from the affected areas, which can thicken over time.[2]

Atopic dermatitis affects about 20% of people at some point in their lives.[2][4] It is more common in younger children.[3] Females are affected slightly more often than males.[6] Millions of people have outgrown the condition.[3]

While the condition may occur at any age, it typically begins in childhood, with varying severity over the years.[2][3] In children under one year of age, the face and limbs and much of the body may be affected.[3] As children get older, the areas on the insides of the knees and folds of the elbows and around the neck are most commonly affected.[3] In adults, the hands and feet are commonly affected.[3] Scratching the affected areas worsens the eczema and increases the risk of skin infections.[2] Many people with atopic dermatitis develop hay fever or asthma.[2] AD commonly begins in early childhood, but it can persist or first appear in adults. It is distinct from, though frequently coexists with, other allergic conditions such as asthma and allergic rhinitis.[7]

The cause is unknown but is believed to involve genetics, immune system dysfunction, environmental exposures, and difficulties with the permeability of the skin.[2][3] If one identical twin is affected, the other has an 85% chance of having the condition.[8] Those who live in cities and dry climates are more commonly affected.[2] Exposure to certain chemicals or frequent hand washing makes symptoms worse.[2] While emotional stress may make the symptoms worse, it is

Atopic dermatitis	
Other names	Atopic eczema, infantile eczema, prurigo Besnier, allergic eczema, neurodermatitis[1]

Atopic dermatitis of the inside crease of the elbow

Specialty	Dermatology, Clinical Immunology and Allergy
Symptoms	Itchy, red, swollen, cracked skin[2]
Complications	Skin infections, hay fever, asthma[2]
Usual onset	Childhood[2][3]
Causes	Unknown[2][3]
Risk factors	Family history, living in a city, dry climate[2]
Diagnostic method	Based on symptoms after ruling out other possible causes[2][3]
Differential diagnosis	Contact dermatitis, psoriasis, seborrheic dermatitis[3]
Treatment	Avoiding things that worsen the condition, daily bathing followed by moisturising cream, steroid creams for flares[3] Humidifier

not a cause.[2] The disorder is not contagious.[2] A diagnosis is typically based on the signs, symptoms, and family history.[3]

Frequency	~20% at some time[2][4]

Treatment involves avoiding things that make the condition worse, enhancing the skin barrier through skin care, and treating the underlying skin inflammation. Moisturising creams are used to make the skin less dry and prevent AD flare-ups. Anti-inflammatory corticosteroid creams are used to control flare-ups.[3] Creams based on calcineurin inhibitors (tacrolimus or pimecrolimus) may also be used to control flares if other measures are not effective.[2][9] Certain antihistamine pills might help with itchiness.[3] Things that commonly make it worse include house dust mite, stress, and seasonal factors.[10] Phototherapy may be useful in some people.[2] Antibiotics (either by mouth or topically) are usually not helpful unless there is secondary bacterial infection or the person is unwell.[11] Dietary exclusion does not benefit most people, and it is only needed if food allergies are suspected.[12] More severe AD cases may need systemic medicines such as cyclosporin, methotrexate, dupilumab, or baricitinib.

Other names of the condition include "infantile eczema", "flexural eczema", "prurigo Besnier", "allergic eczema", and "neurodermatitis".[1]

Signs and symptoms

Symptoms refer to the sensations that people with AD feel, whereas signs refer to a description of the visible changes that result from AD.

Child with atopic dermatitis

The main symptom of AD is itching which can be intense. Some people experience burning, soreness, or pain.[2]

People with AD often have generally dry skin that can look greyish in people with darker skin tones. Areas of AD are not well-defined, and they are typically inflamed (red in light-coloured skin and purple or dark brown in dark-coloured skin).[13] Surface changes include:

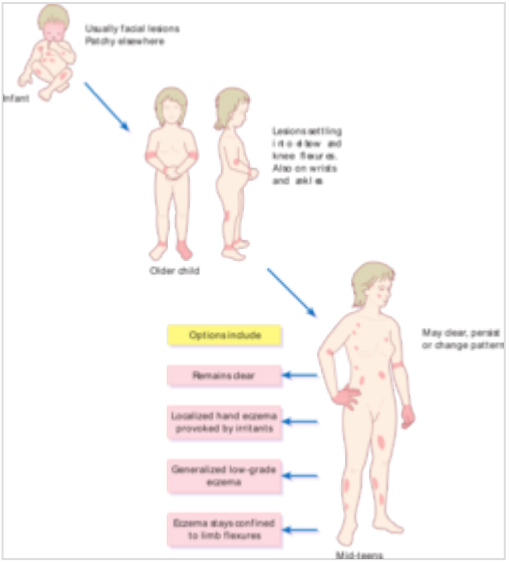
- scaling cracking (skin fissures)
- swelling (oedema)
- scratch marks (excoriation)
- bumpiness (papulation)
- oozing of clear fluid
- thickening of the skin (lichenification) where the AD has been present for a long time.[2]

Eczema often starts on the cheeks and outer limbs and body in infants and frequently settles in the folds of the skin, such as behind the knees, folds of the elbows, around the neck, wrists, and under the buttock folds as the child grows.[14] Any part of the body can be affected by AD.[15]

Atopic dermatitis commonly affects the eyelids, where an extra prominent crease can form under the eyelid due to skin swelling known as Dennie-Morgan infraorbital folds.[16] Cracks can form under the ears, which can be painful (infra-auricular fissure).[17][16]

The inflammation from AD often leaves "footprints" known as postinflammatory pigmentation that can be lighter than the normal skin or darker. These marks are not scars and eventually go back to normal over months, provided the underlying AD is treated effectively.[18]

People with AD often have dry and scaly skin that spans the entire body, except perhaps the diaper area, and intensely itchy red, splotchy, raised lesions that form in the bends of the arms or legs, face, and neck.[19][20][21][22][23]



The pattern of atopic eczema varies with age.

Causes

The cause of AD is not known, although evidence indicates environmental, immunologic, bacterial,[24] and potential genetic factors.[25]

Pollution

Since 1970 the rates of atopic dermatitis in the US and UK have increased three- to sixfold.[26] Even today, people who migrate from developing nations before the age of 4 years old to industrialized nations experience a dramatic rise in the risk of atopic dermatitis and have an additional risk when living in urbanized areas of the industrial nation.[27] Recent work has shed light on these and other data, strongly suggesting that early life industrial exposures may cause atopic dermatitis.[26][28] Chemicals such as (di)isocyanates and xylene prevent the skin bacteria from producing ceramide-sphingolipid family lipids.[26][28] Early life deficiency in these lipids predicts which children will go on to develop atopic dermatitis.[29][30][31][32] These chemicals also directly activate an itch receptor in the skin known as TRPA1.[33] The industrial manufacturing and use of both xylene and diisocyanates greatly increased starting in 1970, which greatly expanded the average exposure to these substances. For example, these chemicals are components of several exposures known to increase the risk of atopic dermatitis or worsen symptoms including: wildfires, automobile exhaust, wallpaper adhesives, paints, non-latex foam furniture, cigarette smoke, and are elements of fabrics like polyester, nylon, and spandex.[27][26][28]

Climate

Low humidity and low temperature increase the prevalence and risk of flare-ups in people with atopic dermatitis.[34]

Genetics

Genes that may contribute to AD are mainly those responsible for immune response (e.g., TH2 cytokine and JAK-STAT pathway genes) and the skin barrier (e.g., filaggrin, claudin-1, loricrin).

Immune response: Many people with AD have a family history or a personal history of atopy. Atopy is a term used to describe individuals who produce substantial amounts of IgE. Such individuals have an increased tendency to develop asthma, hay fever, eczema, urticaria, and allergic rhinitis.[19][20] Up to

80% of people with atopic dermatitis have elevated total or allergen-specific IgE levels.[35]

Skin barrier: About 30% of people with AD have mutations in the gene for the production of filaggrin (*FLG*), which increases the risk for the early onset of atopic dermatitis and developing asthma.[36][37] However, expression of filaggrin protein or breakdown products offers no predictive utility in atopic dermatitis risk.[30]

People with atopic dermatitis also have decreased expression of tight junction protein Claudin-1, which deteriorates the bioelectric barrier function in the epidermis.[38]

Hygiene hypothesis

According to the hygiene hypothesis, early childhood exposure to certain microorganisms (such as gut flora and helminth parasites) protects against allergic diseases by contributing to the development of the immune system.[39] This exposure is limited in a modern "sanitary" environment, and the incorrectly developed immune system is prone to develop allergies to harmless substances.

Some support exists for this hypothesis with respect to AD.[40] Those exposed to dogs while growing up have a lower risk of atopic dermatitis.[41] Also, epidemiological studies support a protective role for helminths against AD.[42] Likewise, children with poor hygiene are at a lower risk for developing AD, as are children who drink unpasteurized milk.[42]

Allergens

In a small percentage of cases, atopic dermatitis is caused by sensitization to foods[43] such as milk, but there is growing consensus that food allergy most likely arises as a result of skin barrier dysfunction resulting from AD, rather than food allergy causing the skin problems.[44] Atopic dermatitis sometimes appears associated with coeliac disease and non-coeliac gluten sensitivity. Because a gluten-free diet (GFD) improves symptoms in these cases, gluten seems to be the cause of AD in these cases.[45][46] A diet high in fruits seems to have a protective effect against AD, whereas the opposite seems true for heavily processed foods.[42]

Exposure to allergens, either from food or the environment, can exacerbate existing atopic dermatitis.[47] Exposure to dust mites, for example, is believed to contribute to the risk of developing AD.[48]

Hard water

The prevalence of atopic dermatitis in children may be linked to the level of calcium carbonate or "hardness" of household drinking water.[49][50] Living in areas with hard water may also play a part in the development of AD in early life. However, when AD is already established, using water softeners at home does not reduce the severity of the symptoms.[50][51]

Role of *Staphylococcus aureus*

Colonization of the skin by the bacterium *S. aureus* is prevalent in those with atopic dermatitis.[52] Abnormalities in the skin barrier of persons with AD are exploited by *S. aureus* to trigger cytokine expression, thus aggravating the condition.[53]

However, atopic dermatitis is non-communicable and therefore could not be directly caused by a highly infectious organism. Furthermore, there is insufficient evidence for the effectiveness of anti-staphylococcal treatments for treating *S. aureus* in infected or uninfected eczema.[54]

The role of *S. aureus* in skin irritation occurs via inflammation factors that induce itching, which may damage the skin, further driving inflammation, and facilitating the growth of *S. aureus*, thus promoting a chronic cycle.[55]

Pathophysiology

Excessive type 2 inflammation underlies the pathophysiology of atopic dermatitis.[56][57]

Disruption of the epidermal barrier is thought to play an integral role in the pathogenesis of AD.[35] Disruptions of the epidermal barrier allows allergens to penetrate the epidermis to deeper layers of the skin. This leads to activation of epidermal inflammatory dendritic and innate lymphoid cells, which subsequently attract Th2 CD4+ helper T cells to the skin.[35] This dysregulated Th2 inflammatory response is thought to lead to the eczematous lesions.[35] The Th2 helper T cells become activated, leading to the release of inflammatory cytokines including IL-4, IL-13, and IL-31 which activate downstream Janus kinase (Jak) pathways. The active Jak pathways lead to inflammation and downstream activation of plasma cells and B lymphocytes, which release antigen-specific IgE, contributing to further inflammation.[35] Other CD4+ helper T-cell pathways thought to be involved in atopic dermatitis inflammation include the Th1, Th17, and Th22 pathways.[35] Some specific CD4+ helper T-cell inflammatory pathways are more commonly activated in specific ethnic groups with AD (for example, the Th-2 and Th-17 pathways are commonly activated in Asian people) possibly explaining the differences in phenotypic presentation of atopic dermatitis in specific populations.[35]

Mutations in the filaggrin gene, *FLG*, also cause impairment in the skin barrier that contributes to the pathogenesis of AD.[35] Filaggrin is produced by epidermal skin cells (keratinocytes) in the horny layer of the epidermis. Filaggrin stimulates skin cells to release moisturizing factors and lipid matrix material, which cause adhesion of adjacent keratinocytes and contribute to the skin barrier.[35] A loss-of-function mutation of filaggrin causes loss of this lipid matrix and external moisturizing factors, subsequently leading to disruption of the skin barrier. The disrupted skin barrier leads to transdermal water loss (leading to the xerosis or dry skin commonly seen in AD) and antigen and allergen penetration of the epidermal layer.[35] Filaggrin mutations are also associated with a decrease in natural antimicrobial peptides found on the skin; subsequently leading to disruption of skin flora and bacterial overgrowth (commonly *Staphylococcus aureus* overgrowth or colonization).[35]

Atopic dermatitis is also associated with the release of pruritogens (molecules that stimulate pruritus or itching) in the skin.[35] Keratinocytes, mast cells, eosinophils and T-cells release pruritogens in the skin; leading to activation of Aδ fibers and Group C nerve fibers in the epidermis and dermis contributing to sensations of pruritus and pain.[35] The pruritogens include the Th2 cytokines IL-4, IL-13, IL-31, histamine, and various neuropeptides.[35] Mechanical stimulation from scratching lesions can also lead to the release of pruritogens contributing to the itch-scratch cycle, whereby there is increased pruritus or itch after scratching a lesion.[35] Chronic scratching of lesions can cause thickening or lichenification of the skin or prurigo nodularis (generalized nodules that are severely itchy).[35]

Another factor in the barrier failure and immunological dysregulation in people with atopic dermatitis may be due to decreases in tight junction protein Claudin-1. Inhibiting Claudin-1 expression in human keratinocytes has been shown to both reduce tight junction function, as well as increase keratinocyte proliferation in vitro. It has also been discovered that this deteriorates the bioelectric barrier function in the epidermis.[38]

Diagnosis

Atopic dermatitis is typically diagnosed clinically, meaning it is based on signs and symptoms alone, without special testing.[58] Several different criteria developed for research have also been validated to aid in diagnosis.[59] Of these, the UK Diagnostic Criteria, based on the work of Hanifin and Rajka, has been the most widely validated.[59][60]

UK diagnostic criteria[60]

People must have itchy skin, or evidence of rubbing or scratching, plus three or more of:
Skin creases are involved – flexural dermatitis of fronts of ankles, antecubital fossae, popliteal fossae, skin around eyes, or neck, (or cheeks for children under 10)
History of asthma or allergic rhinitis (or family history of these conditions if the patient is a child ≤4 years old)
Symptoms began before age 2 (can only be applied to people ≥4 years old)
History of dry skin (within the past year)
Dermatitis is visible on flexural surfaces (people ≥ age 4) or on the cheeks, forehead, and extensor surfaces (people < age 4)

Other diseases that must be excluded before making a diagnosis include contact dermatitis, psoriasis, and seborrheic dermatitis.[3]

Prevention

There are no established clinical methods using dietary or topical strategies to inhibit or prevent atopic dermatitis. Specific dietary plans during pregnancy and in early childhood, such as eating fatty fish (or taking omega-3 supplements), are not effective.[61] Taking probiotics (for example Lactobacillus rhamnosus) during pregnancy and feeding probiotics to infants are strategies under research, with only preliminary evidence that they may be preventative.[62][63]

Using moisturizers daily in infants during the first year of life does not help to prevent atopic dermatitis, and might even increase the risk of skin infections.[51][64]

Treatments

No cure for AD is known, although treatments may reduce the severity and frequency of flares.[19] The most commonly used topical treatments for AD are topical corticosteroids (to get control of flare-ups) and moisturisers (emollients) to help keep control.[65] Clinical trials often measure the efficacy of treatments with a severity scale such as the SCORAD index or the Eczema Area and Severity Index.[58][66]

Moisturisers

Daily basic care stabilizes the barrier function of the skin to mitigate its sensitivity to irritation and penetration of allergens. Affected persons often report that improvement in skin hydration parallels improvement in AD symptoms. Moisturisers (or emollients) can improve skin comfort and may reduce disease flares.[67] They can be used as leave-on treatments, bath additives, or soap substitutes. There are many different products, but the majority of leave-on treatments (least to most greasy) are lotions, creams, gels or ointments. All of the different types of moisturisers are equally effective, so people need to choose one or more products based on what suits them, according to their age, body site affected, climate/season, and personal preference.[68] Non-medicated prescription moisturisers may also be no more effective than over-the-counter moisturisers.[69]

The use of emollient bath additives does not provide any additional benefits.[51][70][71]

Medication

Topical

Creams and ointments containing corticosteroids applied directly on skin (topical) are effective in managing atopic dermatitis.[69][72] Newer (second generation) corticosteroids, such as fluticasone propionate and mometasone furoate, are more effective and safer than older ones. Strong and moderate corticosteroids are more effective than weaker ones. They are also generally safe and do not cause skin thinning when used intermittently to treat AD flare-ups. They are also safe when used twice a week for preventing flares (also known as weekend treatment).[73][69][74] Applying once daily is as effective as twice or more daily application.[72]

In addition to topical corticosteroids, topical calcineurin inhibitors, such as tacrolimus or pimecrolimus, are also recommended as first-line therapies for managing atopic dermatitis.[69][75] Both tacrolimus and pimecrolimus are effective and safe to use in AD.[76][77] Crisaborole, an inhibitor of PDE-4, is also effective and safe as a topical treatment for mild-to-moderate AD.[78][79] Ruxolitinib, a Janus kinase inhibitor, has uncertain efficacy and safety.[69][75]

Difamilast, a PDE4 inhibitor, was approved for medical use in Japan in September 2021,[80] and in the United States in February 2026.[81]

Systemic

When topical (on skin) treatments fail to control severe AD flares, medications taken by mouth (systemic treatment) can be used.[51]

Conventional oral medications for AD include systemic immunosuppressants, such as ciclosporin, methotrexate, azathioprine, and mycophenolate.[82][83][84][51] Antidepressants and antihistamine may be used to control pruritus (itchiness).[85]

Newer medications, such as monoclonal antibodies and JAK inhibitors, are highly effective for managing atopic dermatitis, but modestly increase the risk of conjunctivitis. These include dupilumab (Dupixent), tralokinumab (Adtralza, Adbry), abrocitinib (Cibinqo), baricitinib (Olumiant) and upadacitinib (Rinvoq).[82][75][86] Among monoclonal antibodies, dupilumab and tralokinumab are approved to treat

moderate-to-severe eczema in the US and the EU.[87][88][89][90] Lebrikizumab is also approved in the EU for treating moderate-to-severe AD[91] but in the US its approval was declined due to manufacturing issues.[92] Abrocitinib and upadacitinib have also been approved in the US for the treatment of moderate-to-severe eczema.[93][94] Nemolizumab (Nemluvio) was approved to treat atopic dermatitis in December 2024.[95]

Allergen immunotherapy may be effective in relieving symptoms of AD, but it also comes with an increased risk of adverse events.[96] This treatment consists of a series of injections or drops under the tongue of a solution containing the allergen.[97]

The skin of people with AD can easily get infected, most commonly by the bacteria Staphylococcus aureus. Signs of this include oozing fluid, a yellow crust on the skin, worsening eczema symptoms, and fever. Antibiotics are commonly used to target overgrowth of *S. aureus*, but their benefit is limited, and they increase the risk of antimicrobial resistance. For these reasons, they are only recommended for people who not only present symptoms on the skin but also feel systemically unwell.[51][54][98]

Diet

The role of vitamin D on atopic dermatitis is not clear, but vitamin D supplementation may improve its symptoms.[99][100][101]

There is no clear benefit for pregnant mothers taking omega 3 long-chain polyunsaturated fatty acid (LCPUFA) in preventing the development of AD in their child.[102][103]

Several probiotics seem to have a positive effect, with a roughly 20% reduction in the rate of AD.[104][105][106] Probiotics containing multiple strains of bacteria seem to work the best.[107]

In people with celiac disease or nonceliac gluten sensitivity, a gluten-free diet improves their symptoms and prevents the occurrence of new outbreaks.[45][46]

Use of blood-specific IgE or skin prick tests to guide dietary exclusions to improve disease severity or control is controversial. Clinicians vary in their use of these tests for this purpose, and there is very limited evidence of any benefit.[108]

Lifestyle

Health professionals often recommend that people with AD bathe regularly in lukewarm baths, especially in saltwater, to moisten their skin.[20][109] Dilute bleach baths may be helpful for people with moderate and severe eczema, but only for people with Staphylococcus aureus.[110]

Avoiding large-diameter woolen clothing or scratchy fibres is usually recommended for people with AD as they can trigger a flare.[111][112] Safe alternatives are clothes made from fabrics with smaller diameters and smooth fibers. These include super- and ultrafine merino wool and fabrics with anti-microbial textile finishes. Wearing silk is also safe but does not improve symptoms of AD.[111][51][113]

Self-management

Living with AD requires a high level of self-management (for example, avoiding triggers) and adherence to treatments (regularly applying medication). Good self-management contributes to better disease outcomes and quality of life.[114][115] However, worries about topical treatments, misconceptions about the condition, unclear information, and unsuitable communication from doctors can make living with AD more difficult.[51]

People with AD often do not regard eczema as a long-term condition and hope they will outgrow or cure it. This can cause worse adherence to the necessary long-term treatment. Doctors should not imply that it is a short-term condition and should emphasise that even though it cannot be cured, it can be controlled effectively.[116][115]

Appropriate communication from doctors can support self-management. Doctors need to address concerns about treatments and provide clear and consistent information about the condition.[116][115] Treatment regimens can be confusing, and written action plans may support people in knowing which treatments to use where and when.[117] A website supporting self-management has been shown to improve AD symptoms for parents, children, adolescents and young adults.[118][119]

Light

Phototherapic treatment involves exposure to broad- or narrow-band ultraviolet (UV) light. UV radiation exposure has been found to have a localized immunomodulatory effect on affected tissues and may be used to decrease the severity and frequency of flares.[120][121] Among the different types of phototherapies only narrowband (NB) ultraviolet B (UVB) exposure might help with the severity of AD and ease itching.[86][122] However, UV radiation has also been implicated in various types of skin cancer, and thus UV treatment is not without risk.[123] UV phototherapy is not indicated in young adults and children due to this risk of skin cancer with prolonged use or exposure.[35]

Alternative medicine

While several Chinese herbal medicines are intended for treating atopic eczema, there is no evidence showing that these treatments, taken by mouth or applied topically, reduce the severity of eczema in children or adults.[124] Research into microbiome-based approaches to restoring the skin barrier has also been explored, including formulations developed by companies such as Codex Labs, which focus on supporting skin microbiota balance in atopic dermatitis.[125]

Impact

Atopic dermatitis significantly impairs the quality of life of affected individuals.[126][127] The impact of AD extends beyond physical symptoms, encompassing substantial humanistic and psychosocial effects. Its burden is significant, especially given the high indirect costs and psychological impacts on quality of life.[126][128]

According to the Global Burden of Disease Study, AD is the skin disease with the highest disability-adjusted life year burden and ranks in the top 15 of all nonfatal diseases. In comparison with other dermatological conditions like psoriasis and urticaria, AD presents a significantly higher burden.[127]

While AD remains incurable, reducing its severity can significantly alleviate its burden. Understanding the extent of the burden of AD can aid in better resource allocation and prioritization of interventions, benefiting both people with atopic dermatitis and healthcare systems.[129]

Humanistic burden

Atopic dermatitis significantly decreases the quality of life by affecting various aspects of people's lives. The psychological impact, often resulting in conditions like depression and anxiety, is a major factor leading to decreased quality of life. Sleep disturbances, commonly reported in people with AD, further contribute to the humanistic burden, affecting daily productivity and concentration.[126]

Clinical and economic burden

Economically, AD imposes a substantial burden on healthcare systems, with the average direct cost per patient estimated at US $4411 and the average indirect cost reaching US $9068 annually.[126] These figures highlight the considerable financial impact of the disease on healthcare systems and people with the condition.[130][131]

Productivity loss

Atopic dermatitis also has a marked impact on productivity. The total number of days lost annually due to these factors is about 68.8 days for the general AD population, with presenteeism accounting for the majority of these days.[126] The impact on productivity varies significantly with the severity of AD, with more severe cases resulting in higher numbers of days lost.[126][132]

Burden of disease in the Middle East and Africa

Atopic dermatitis leads to the highest loss in disability-adjusted life years compared to other skin diseases in the Middle East and Africa.[133] Patients with AD in these regions lose approximately 0.19 quality-adjusted life years (QALYs) annually due to the disease. Egypt experiences the highest QALY loss and Kuwait the lowest.[133]

The average annual healthcare cost per patient varies; is highest in the United Arab Emirates, estimated at US $3569, and lowest in Algeria at US $312. These costs are influenced by the economic status of each country and the cost of healthcare. Advanced treatments like targeted therapies and phototherapy are among the main cost drivers.[133]

Indirect costs, primarily due to productivity loss from absenteeism and presenteeism, average about 67% in these countries. Indirect costs in Saudi Arabia are the highest in the area, estimated at US $364 million.[133] Factors like mental health impact, side effects of treatments, and other indirect costs, such as personal care products, are not fully accounted for in these estimates, suggesting that the actual burden might be even higher.[133]

To mitigate the burden of AD, experts recommend strategic actions across five key domains: capacity building, guidelines, research, public awareness, and patient support and education. Key measures include increasing the number of dermatologists, establishing evidence-based treatment guidelines,

investing in patient education, and enhancing public awareness to reduce stigma.[134] Improving access to effective treatments and conducting further research on AD's impact are also crucial for reducing the disease's clinical, economic, and humanistic burdens in the MEA.[134]

Epidemiology

Since the beginning of the 20th century, many inflammatory skin disorders have become more common; AD is a classic example of such a disease. Although AD was previously considered primarily a childhood disease, it is now recognized as highly prevalent in adults, with an estimated adult prevalence of 3–5% globally.[126][127] It now affects 15–30% of children and 2–10% of adults in developed countries, and in the United States has nearly tripled in the past 30–40 years.[20][135] Over 15 million American adults and children have AD.[136]

Society and culture

Conspiracy theories

A number of false and conspiratorial claims about AD have emerged on the internet and have been amplified by social media. These conspiracy theories include, among others, claims that AD is caused by 5G, formaldehyde in food, vaccines, and topical steroids. Various unproven theories also claim that vegan diets, apple cider vinegar, calendula, and witch hazel can cure AD and that air purifiers reduce the risk of developing AD.[137]

Research

Leukotriene receptor antagonists, such as montelukast, might be useful for the treatment of AD but their effectiveness has not yet been proven by research.[138][139][140]

References

1. Williams HC (October 2000). "Epidemiology of atopic dermatitis" (https://books.google.com/books?id=q8OZ4O_gjQUC&pg=PA10). *Clinical and Experimental Dermatology*. **25** (7). Cambridge University Press: 522–529. doi:10.1046/j.1365-2230.2000.00698.x (https://doi.org/10.1046%2Fj.1365-2230.2000.00698.x). ISBN 978-0-521-57075-6. PMID 11122223 (https://pubmed.ncbi.nlm.nih.gov/11122223). S2CID 31546363 (https://api.semanticscholar.org/CorpusID:31546363). Archived (https://web.archive.org/web/20150619134904/https://books.google.ca/books?id=q8OZ4O_gjQUC&pg=PA10) from the original on 19 June 2015.
2. "Handout on Health: Atopic Dermatitis (A type of eczema)" (https://www.niams.nih.gov/health-topics/atopic-dermatitis). *National Institute of Arthritis and Musculoskeletal and Skin Diseases*. May 2013. Archived (https://web.archive.org/web/20150530092344/http://www.niams.nih.gov/Health_Info/Atopic_Dermatitis/default.asp) from the original on 30 May 2015. Retrieved 19 June 2015.

3. Tollefson MM, Bruckner AL (December 2014). "Atopic dermatitis: skin-directed management" (https://doi.org/10.1542%2Fpeds.2014-2812). *Pediatrics*. **134** (6): e1735–e1744. doi:10.1542/peds.2014-2812 (https://doi.org/10.1542%2Fpeds.2014-2812). PMID 25422009 (https://pubmed.ncbi.nlm.nih.gov/25422009).
4. Thomsen SF (2014). "Atopic dermatitis: natural history, diagnosis, and treatment" (https://doi.org/10.1155%2F2014%2F354250). *ISRN Allergy*. **2014** 354250. doi:10.1155/2014/354250 (https://doi.org/10.1155%2F2014%2F354250). PMC 4004110 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4004110). PMID 25006501 (https://pubmed.ncbi.nlm.nih.gov/25006501).
5. Johansson SG, Bieber T, Dahl R, Friedmann PS, Lanier BQ, Lockey RF, et al. (May 2004). "Revised nomenclature for allergy for global use: Report of the Nomenclature Review Committee of the World Allergy Organization, October 2003" (https://doi.org/10.1016%2Fj.jaci.2003.12.591). *The Journal of Allergy and Clinical Immunology*. **113** (5): 832–836. Bibcode:2004JACI..113..832J (https://ui.adsabs.harvard.edu/abs/2004JACI..113..832J). doi:10.1016/j.jaci.2003.12.591 (https://doi.org/10.1016%2Fj.jaci.2003.12.591). PMID 15131563 (https://pubmed.ncbi.nlm.nih.gov/15131563).
6. "Atopic Dermatitis" (https://www.niams.nih.gov/health-topics/atopic-dermatitis). *National Institute of Arthritis and Musculoskeletal and Skin Diseases*. September 2019. Retrieved 29 August 2022.
7. "Atopic Dermatitis" (https://phoilex.com/blogs/glossary/atopic-dermatitis). *Phoilex*. Retrieved 23 March 2026.
8. Williams H (2009). *Evidence-Based Dermatology* (https://books.google.com/books?id=SbsQij5xkfYC&pg=PA128). John Wiley & Sons. p. 128. ISBN 978-1-4443-0017-8. Archived (https://web.archive.org/web/20170908181020/https://books.google.com/books?id=SbsQij5xkfYC&pg=PA128) from the original on 8 September 2017.
9. Carr WW (August 2013). "Topical calcineurin inhibitors for atopic dermatitis: review and treatment recommendations" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3715696). *Paediatric Drugs*. **15** (4): 303–310. doi:10.1007/s40272-013-0013-9 (https://doi.org/10.1007%2Fs40272-013-0013-9). PMC 3715696 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3715696). PMID 23549982 (https://pubmed.ncbi.nlm.nih.gov/23549982).
10. Langan SM, Williams HC (September 2006). "What causes worsening of eczema? A systematic review". *The British Journal of Dermatology*. **155** (3): 504–514. doi:10.1111/j.1365-2133.2006.07381.x (https://doi.org/10.1111%2Fj.1365-2133.2006.07381.x). PMID 16911274 (https://pubmed.ncbi.nlm.nih.gov/16911274). S2CID 43247714 (https://api.semanticscholar.org/CorpusID:43247714).
11. Ong PY, Boguniewicz J, Chu DK (May 2023). "Skin Antiseptics for Atopic Dermatitis: Dissecting Facts From Fiction". *The Journal of Allergy and Clinical Immunology. In Practice*. **11** (5): 1385–1390. doi:10.1016/j.jaip.2023.01.012 (https://doi.org/10.1016%2Fj.jaip.2023.01.012). PMID 36702247 (https://pubmed.ncbi.nlm.nih.gov/36702247). S2CID 256222372 (https://api.semanticscholar.org/CorpusID:256222372).
12. Oykhman P, Dookie J, Al-Rammahy H, de Benedetto A, Asiniwasis RN, LeBovidge J, et al. (October 2022). "Dietary Elimination for the Treatment of Atopic Dermatitis: A Systematic Review and Meta-Analysis". *The Journal of Allergy and Clinical Immunology. In Practice*. **10** (10): 2657–2666.e8. doi:10.1016/j.jaip.2022.06.044 (https://doi.org/10.1016%2Fj.jaip.2022.06.044). PMID 35987995 (https://pubmed.ncbi.nlm.nih.gov/35987995). S2CID 250710625 (https://api.semanticscholar.org/CorpusID:250710625).
13. Kaufman BP, Guttman-Yassky E, Alexis AF (April 2018). "Atopic dermatitis in diverse racial and ethnic groups-Variations in epidemiology, genetics, clinical presentation and treatment" (https://doi.org/10.1111%2Fexd.13514). *Experimental Dermatology*. **27** (4): 340–357. doi:10.1111/exd.13514 (https://doi.org/10.1111%2Fexd.13514). PMID 29457272 (https://pubmed.ncbi.nlm.nih.gov/29457272). S2CID 3379280 (https://api.semanticscholar.org/CorpusID:3379280).

14. Rook A, Burns T (2010). "Atopic dermatitis". *Rook's Textbook of Dermatology* (8th ed.). Chichester, UK: Wiley-Blackwell. ISBN 978-1-4443-1764-0. OCLC 605909001 (https://search.worldcat.org/oclc/605909001).
15. Friedmann PS, Holden CA (January 2004). "Atopic Dermatitis". In Burns T, Breathnach S, Cox N, Griffiths C (eds.). *Rook's Textbook of Dermatology*. Malden, Massachusetts, USA: Blackwell Publishing, Inc. pp. 755–786. doi:10.1002/9780470750520.ch18 (https://doi.org/10.1002%2F9780470750520.ch18). ISBN 978-0-470-75052-0.
16. Kwatra SG, Tey HL, Ali SM, Dabade T, Chan YH, Yosipovitch G (June 2012). "The infra-auricular fissure: a bedside marker of disease severity in patients with atopic dermatitis". *Journal of the American Academy of Dermatology*. **66** (6): 1009–1010. doi:10.1016/j.jaad.2011.10.031 (https://doi.org/10.1016%2Fj.jaad.2011.10.031). PMID 22583715 (https://pubmed.ncbi.nlm.nih.gov/22583715).
17. Langan SM, Irvine AD, Weidinger S (August 2020). "Atopic dermatitis". *Lancet*. **396** (10247): 345–360. Bibcode:2020Lanc..396..345L (https://ui.adsabs.harvard.edu/abs/2020Lanc..396..345L). doi:10.1016/S0140-6736(20)31286-1 (https://doi.org/10.1016%2FS0140-6736%2820%2931286-1). PMID 32738956 (https://pubmed.ncbi.nlm.nih.gov/32738956). S2CID 220873055 (https://api.semanticscholar.org/CorpusID:220873055).
18. Lambert A (9 February 2021). "Skin pigmentation and eczema" (https://eczema.org/information-and-advice/living-with-eczema/skin-pigmentation/). *National Eczema Society*. Retrieved 6 April 2023.
19. Berke R, Singh A, Guralnick M (July 2012). "Atopic dermatitis: an overview" (http://www.aafp.org/afp/2012/0701/p35.pdf) (PDF). *American Family Physician*. **86** (1): 35–42. PMID 22962911 (https://pubmed.ncbi.nlm.nih.gov/22962911). Archived (https://web.archive.org/web/20150906111528/http://www.aafp.org/afp/2012/0701/p35.pdf) (PDF) from the original on 6 September 2015.
20. Kim BS (21 January 2014). Fritsch P, Vinson RP, Perry V, Quirk CM, James WD (eds.). "Atopic Dermatitis" (http://emedicine.medscape.com/article/1049085-overview#showall). *Medscape Reference*. WebMD. Archived (https://web.archive.org/web/20140210011334/http://emedicine.medscape.com/article/1049085-overview#showall) from the original on 10 February 2014. Retrieved 3 March 2014.
21. Brehler R (2009). "Atopic Dermatitis". In Lang F (ed.). *Encyclopedia of molecular mechanisms of diseases*. Berlin: Springer. ISBN 978-3-540-67136-7.
22. Baron SE, Cohen SN, Archer CB (May 2012). "Guidance on the diagnosis and clinical management of atopic eczema" (https://doi.org/10.1111%2Fj.1365-2230.2012.04336.x). *Clinical and Experimental Dermatology*. **37** (Suppl 1): 7–12. doi:10.1111/j.1365-2230.2012.04336.x (https://doi.org/10.1111%2Fj.1365-2230.2012.04336.x). PMID 22486763 (https://pubmed.ncbi.nlm.nih.gov/22486763). S2CID 28538214 (https://api.semanticscholar.org/CorpusID:28538214).
23. Schmitt J, Langan S, Deckert S, Svensson A, von Kobyletzki L, Thomas K, et al. (December 2013). "Assessment of clinical signs of atopic dermatitis: a systematic review and recommendation" (https://doi.org/10.1016%2Fj.jaci.2013.07.008). *The Journal of Allergy and Clinical Immunology*. **132** (6): 1337–1347. doi:10.1016/j.jaci.2013.07.008 (https://doi.org/10.1016%2Fj.jaci.2013.07.008). PMID 24035157 (https://pubmed.ncbi.nlm.nih.gov/24035157).
24. Kedzierska A, Kapińska-Mrowiecka M, Czubak-Macugowska M, Wójcik K, Kedzierska J (December 2008). "Susceptibility testing and resistance phenotype detection in Staphylococcus aureus strains isolated from patients with atopic dermatitis, with apparent and recurrent skin colonization" (https://ruj.uj.edu.pl/xmlui/handle/item/76783). *The British Journal of Dermatology*. **159** (6): 1290–1299. doi:10.1111/j.1365-2133.2008.08817.x (https://doi.org/10.1111%2Fj.1365-2133.2008.08817.x). ISSN 1365-2133 (https://search.worldcat.org/issn/1365-2133). PMID 18795934 (https://pubmed.ncbi.nlm.nih.gov/18795934).

25. Grey K, Maguiness S (August 2016). "Atopic Dermatitis: Update for Pediatricians". *Pediatric Annals* (Review). **45** (8): e280–e286. doi:10.3928/19382359-20160720-05 (https://doi.org/10.3928%2F19382359-20160720-05). PMID 27517355 (https://pubmed.ncbi.nlm.nih.gov/27517355).
26. "NIAID Researchers Identify Link Between Common Chemicals and Eczema | NIH: National Institute of Allergy and Infectious Diseases" (https://www.niaid.nih.gov/news-events/researchers-identify-link-between-common-chemicals-and-eczema). *www.niaid.nih.gov*. 23 January 2023. Retrieved 23 November 2023.
27. Zeldin J, Chaudhary PP, Spathies J, Yadav M, D'Souza BN, Alishahedani ME, et al. (January 2023). "Exposure to isocyanates predicts atopic dermatitis prevalence and disrupts therapeutic pathways in commensal bacteria" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9821876). *Science Advances*. **9** (1) eade8898. Bibcode:2023SciA....9E8898Z (https://ui.adsabs.harvard.edu/abs/2023SciA....9E8898Z). doi:10.1126/sciadv.ade8898 (https://doi.org/10.1126%2Fsciadv.ade8898). PMC 9821876 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9821876). PMID 36608129 (https://pubmed.ncbi.nlm.nih.gov/36608129).
28. "Eczema's cause could be in the air we breathe" (https://www.nbcnews.com/health/health-news/causes-eczema-air-breathe-new-science-suggests-rcna76427). *NBC News*. 26 March 2023. Retrieved 23 November 2023.
29. Chaudhary PP, Myles IA, Zeldin J, Dabdoub S, Deopujari V, Baveja R, et al. (October 2023). "Shotgun metagenomic sequencing on skin microbiome indicates dysbiosis exists prior to the onset of atopic dermatitis" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10543534). *Allergy*. **78** (10): 2724–2731. doi:10.1111/all.15806 (https://doi.org/10.1111%2Fall.15806). PMC 10543534 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10543534). PMID 37422700 (https://pubmed.ncbi.nlm.nih.gov/37422700).
30. Berdyshev E, Kim J, Kim BE, Goleva E, Lyubchenko T, Bronova I, et al. (May 2023). "Stratum corneum lipid and cytokine biomarkers at age 2 months predict the future onset of atopic dermatitis". *The Journal of Allergy and Clinical Immunology*. **151** (5): 1307–1316. doi:10.1016/j.jaci.2023.02.013 (https://doi.org/10.1016%2Fj.jaci.2023.02.013). PMID 36828081 (https://pubmed.ncbi.nlm.nih.gov/36828081). S2CID 255078763 (https://api.semanticscholar.org/CorpusID:255078763).
31. Rinnov MR, Halling AS, Gerner T, Ravn NH, Knudgaard MH, Trautner S, et al. (March 2023). "Skin biomarkers predict development of atopic dermatitis in infancy" (https://doi.org/10.1111%2Fall.15518). *Allergy*. **78** (3): 791–802. doi:10.1111/all.15518 (https://doi.org/10.1111%2Fall.15518). PMID 36112082 (https://pubmed.ncbi.nlm.nih.gov/36112082).
32. Yamamoto-Hanada K, Saito-Abe M, Shima K, Fukagawa S, Uehara Y, Ueda Y, et al. (July 2023). "mRNAs in skin surface lipids unveiled atopic dermatitis at 1 month" (https://doi.org/10.1111%2Fjdv.19017). *Journal of the European Academy of Dermatology and Venereology*. **37** (7): 1385–1395. doi:10.1111/jdv.19017 (https://doi.org/10.1111%2Fjdv.19017). PMID 36897437 (https://pubmed.ncbi.nlm.nih.gov/36897437).
33. Yadav M, Chaudhary PP, D'Souza BN, Ratley G, Spathies J, Ganesan S, et al. (2023). "Diisocyanates influence models of atopic dermatitis through direct activation of TRPA1" (https://doi.org/10.1371%2Fjournal.pone.0282569). *PLOS ONE*. **18** (3) e0282569. Bibcode:2023PLoSO..1882569Y (https://ui.adsabs.harvard.edu/abs/2023PLoSO..1882569Y). doi:10.1371/journal.pone.0282569 (https://doi.org/10.1371%2Fjournal.pone.0282569). PMC 9987805 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9987805). PMID 36877675 (https://pubmed.ncbi.nlm.nih.gov/36877675).
34. Engebretsen KA, Johansen JD, Kezic S, Linneberg A, Thyssen JP (February 2016). "The effect of environmental humidity and temperature on skin barrier function and dermatitis" (https://doi.org/10.1111%2Fjdv.13301). *Journal of the European Academy of Dermatology and Venereology*. **30** (2): 223–249. doi:10.1111/jdv.13301 (https://doi.org/10.1111%2Fjdv.13301). PMID 26449379 (https://pubmed.ncbi.nlm.nih.gov/26449379). S2CID 12378072 (https://api.semanticscholar.org/CorpusID:12378072).

35. Ständer S (March 2021). "Atopic Dermatitis". *The New England Journal of Medicine*. **384** (12): 1136–1143. doi:10.1056/NEJMra2023911 (https://doi.org/10.1056%2FNEJMra2023911). PMID 33761208 (https://pubmed.ncbi.nlm.nih.gov/33761208). S2CID 232355341 (https://api.semanticscholar.org/CorpusID:232355341).
36. Park KD, Pak SC, Park KK (December 2016). "The Pathogenetic Effect of Natural and Bacterial Toxins on Atopic Dermatitis" (https://doi.org/10.3390%2Ftoxins9010003). *Toxins* (Review). **9** (1): 3. doi:10.3390/toxins9010003 (https://doi.org/10.3390%2Ftoxins9010003). PMC 5299398 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5299398). PMID 28025545 (https://pubmed.ncbi.nlm.nih.gov/28025545).
37. Irvine AD, McLean WH, Leung DY (October 2011). "Filaggrin mutations associated with skin and allergic diseases". *The New England Journal of Medicine* (Review). **365** (14): 1315–1327. doi:10.1056/NEJMra1011040 (https://doi.org/10.1056%2FNEJMra1011040). PMID 21991953 (https://pubmed.ncbi.nlm.nih.gov/21991953).
38. Benedetto De A (2010). "Tight Junction Defects in Atopic Dermatitis" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3049863). *Journal of Allergy and Clinical Immunology*. **127** (3): 773–786. doi:10.1016/j.jaci.2010.10.018 (https://doi.org/10.1016%2Fj.jaci.2010.10.018). PMC 3049863 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3049863). PMID 21163515 (https://pubmed.ncbi.nlm.nih.gov/21163515).
39. Scudellari M (February 2017). "News Feature: Cleaning up the hygiene hypothesis" (https://doi.org/10.1073%2Fpnas.1700688114). *Proceedings of the National Academy of Sciences of the United States of America*. **114** (7): 1433–1436. Bibcode:2017PNAS..114.1433S (https://ui.adsabs.harvard.edu/abs/2017PNAS..114.1433S). doi:10.1073/pnas.1700688114 (https://doi.org/10.1073%2Fpnas.1700688114). PMC 5320962 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5320962). PMID 28196925 (https://pubmed.ncbi.nlm.nih.gov/28196925).
40. Bieber T (April 2008). "Atopic dermatitis". *The New England Journal of Medicine*. **358** (14): 1483–1494. doi:10.1056/NEJMra074081 (https://doi.org/10.1056%2FNEJMra074081). PMID 18385500 (https://pubmed.ncbi.nlm.nih.gov/18385500).
41. Pelucchi C, Galeone C, Bach JF, La Vecchia C, Chatenoud L (September 2013). "Pet exposure and risk of atopic dermatitis at the pediatric age: a meta-analysis of birth cohort studies". *The Journal of Allergy and Clinical Immunology*. **132** (3): 616–622.e7. doi:10.1016/j.jaci.2013.04.009 (https://doi.org/10.1016%2Fj.jaci.2013.04.009). hdl:2434/239729 (https://hdl.handle.net/2434%2F239729). PMID 23711545 (https://pubmed.ncbi.nlm.nih.gov/23711545).
42. Flohr C, Mann J (January 2014). "New insights into the epidemiology of childhood atopic dermatitis" (https://doi.org/10.1111%2Fall.12270). *Allergy*. **69** (1): 3–16. doi:10.1111/all.12270 (https://doi.org/10.1111%2Fall.12270). PMID 24417229 (https://pubmed.ncbi.nlm.nih.gov/24417229). S2CID 32645590 (https://api.semanticscholar.org/CorpusID:32645590).
43. di Mauro G, Bernardini R, Barberi S, Capuano A, Correra A, De' Angelis GL, et al. (2016). "Prevention of food and airway allergy: consensus of the Italian Society of Preventive and Social Paediatrics, the Italian Society of Paediatric Allergy and Immunology, and Italian Society of Pediatrics" (https://doi.org/10.1186%2Fs40413-016-0111-6). *The World Allergy Organization Journal* (Review). **9** 28. doi:10.1186/s40413-016-0111-6 (https://doi.org/10.1186%2Fs40413-016-0111-6). PMC 4989298 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4989298). PMID 27583103 (https://pubmed.ncbi.nlm.nih.gov/27583103).
44. Brough HA, Nadeau KC, Sindher SB, Alkotob SS, Chan S, Bahnson HT, et al. (September 2020). "Epicutaneous sensitization in the development of food allergy: What is the evidence and how can this be prevented?" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7494573). *Allergy*. **75** (9): 2185–2205. doi:10.1111/all.14304 (https://doi.org/10.1111%2Fall.14304). PMC 7494573 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7494573). PMID 32249942 (https://pubmed.ncbi.nlm.nih.gov/32249942).

45. Fasano A, Sapone A, Zevallos V, Schuppan D (May 2015). "Nonceliac gluten sensitivity" (https://doi.org/10.1053%2Fj.gastro.2014.12.049). *Gastroenterology* (Review). **148** (6): 1195–1204. doi:10.1053/j.gastro.2014.12.049 (https://doi.org/10.1053%2Fj.gastro.2014.12.049). PMID 25583468 (https://pubmed.ncbi.nlm.nih.gov/25583468). "Many patients with celiac disease also have atopic disorders. About 30% of patients' allergies with gastrointestinal (GI) symptoms and mucosal lesions, but negative results from serologic (TG2 antibodies) or genetic tests (DQ2 or DQ8 genotype) for celiac disease, had reduced GI and atopic symptoms when they were placed on GFDs. These findings indicated that their symptoms were related to gluten ingestion."
46. Mansueto P, Seidita A, D'Alcamo A, Carroccio A (2014). "Non-celiac gluten sensitivity: literature review". *Journal of the American College of Nutrition* (Review). **33** (1): 39–54. doi:10.1080/07315724.2014.869996 (https://doi.org/10.1080%2F07315724.2014.869996). hdl:10447/90208 (https://hdl.handle.net/10447%2F90208). PMID 24533607 (https://pubmed.ncbi.nlm.nih.gov/24533607). S2CID 22521576 (https://api.semanticscholar.org/CorpusID:22521576).
47. Williams H, Flohr C (July 2006). "How epidemiology has challenged 3 prevailing concepts about atopic dermatitis" (https://web.archive.org/web/20180719092709/http://eprints.nottingham.ac.uk/861/2/revised_final_rostrum.pdf) (PDF). *The Journal of Allergy and Clinical Immunology*. **118** (1): 209–213. doi:10.1016/j.jaci.2006.04.043 (https://doi.org/10.1016%2Fj.jaci.2006.04.043). PMID 16815157 (https://pubmed.ncbi.nlm.nih.gov/16815157). Archived from the original (http://eprints.nottingham.ac.uk/861/2/revised_final_rostrum.pdf) (PDF) on 19 July 2018. Retrieved 5 February 2019.
48. Fuiano N, Incorvaia C (June 2012). "Dissecting the causes of atopic dermatitis in children: less foods, more mites" (https://doi.org/10.2332%2Fallergolint.11-RA-0371). *Allergology International*. **61** (2): 231–243. doi:10.2332/allergolint.11-RA-0371 (https://doi.org/10.2332%2Fallergolint.11-RA-0371). PMID 22361514 (https://pubmed.ncbi.nlm.nih.gov/22361514).
49. Sengupta P (August 2013). "Potential health impacts of hard water" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3775162). *International Journal of Preventive Medicine* (Review). **4** (8): 866–875. PMC 3775162 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3775162). PMID 24049611 (https://pubmed.ncbi.nlm.nih.gov/24049611).
50. Jabbar-Lopez ZK, Ung CY, Alexander H, Gurung N, Chalmers J, Danby S, et al. (March 2021). "The effect of water hardness on atopic eczema, skin barrier function: A systematic review, meta-analysis". *Clinical and Experimental Allergy*. **51** (3): 430–451. doi:10.1111/cea.13797 (https://doi.org/10.1111%2Fcea.13797). PMID 33259122 (https://pubmed.ncbi.nlm.nih.gov/33259122). S2CID 227245344 (https://api.semanticscholar.org/CorpusID:227245344).
51. Eczema in children: uncertainties addressed (https://evidence.nihr.ac.uk/collection/eczema-in-children-uncertainties-addressed/) (Report). NIHR Evidence. 19 March 2024. doi:10.3310/nihrevidence_62438 (https://doi.org/10.3310%2Fnihrevidence_62438).
52. Goh CL, Wong JS, Giam YC (September 1997). "Skin colonization of *Staphylococcus aureus* in atopic dermatitis patients seen at the National Skin Centre, Singapore". *International Journal of Dermatology*. **36** (9): 653–657. doi:10.1046/j.1365-4362.1997.00290.x (https://doi.org/10.1046%2Fj.1365-4362.1997.00290.x). PMID 9352404 (https://pubmed.ncbi.nlm.nih.gov/9352404). S2CID 3112669 (https://api.semanticscholar.org/CorpusID:3112669).
53. Nakatsuji T, Chen TH, Two AM, Chun KA, Narala S, Geha RS, et al. (November 2016). "Staphylococcus aureus Exploits Epidermal Barrier Defects in Atopic Dermatitis to Trigger Cytokine Expression" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5103312). *The Journal of Investigative Dermatology*. **136** (11): 2192–2200. doi:10.1016/j.jid.2016.05.127 (https://doi.org/10.1016%2Fj.jid.2016.05.127). PMC 5103312 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5103312). PMID 27381887 (https://pubmed.ncbi.nlm.nih.gov/27381887).

54. George SM, Karanovic S, Harrison DA, Rani A, Birnie AJ, Bath-Hextall FJ, et al. (Cochrane Skin Group) (October 2019). "Interventions to reduce Staphylococcus aureus in the management of eczema" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6818407). *The Cochrane Database of Systematic Reviews*. **2019** (10). doi:10.1002/14651858.CD003871.pub3 (https://doi.org/10.1002%2F14651858.CD003871.pub3). PMC 6818407 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6818407). PMID 31684694 (https://pubmed.ncbi.nlm.nih.gov/31684694).
55. Gallo RL, Horswill AR (May 2024). "Staphylococcus aureus: The Bug Behind the Itch in Atopic Dermatitis" (https://doi.org/10.1016%2Fj.jid.2024.01.001). *Journal of Investigative Dermatology*. **144** (5): 950–953. doi:10.1016/j.jid.2024.01.001 (https://doi.org/10.1016%2Fj.jid.2024.01.001). PMID 38430083 (https://pubmed.ncbi.nlm.nih.gov/38430083). *This article incorporates text from this source, which is available under the CC BY 4.0 license.*
56. Gandhi NA, Bennett BL, Graham NM, Pirozzi G, Stahl N, Yancopoulos GD (January 2016). "Targeting key proximal drivers of type 2 inflammation in disease". *Nature Reviews. Drug Discovery*. **15** (1): 35–50. Bibcode:2016NRvDD..15...35G (https://ui.adsabs.harvard.edu/abs/2016NRvDD..15...35G). doi:10.1038/nrd4624 (https://doi.org/10.1038%2Fnrd4624). PMID 26471366 (https://pubmed.ncbi.nlm.nih.gov/26471366). S2CID 2421591 (https://api.semanticscholar.org/CorpusID:2421591).
57. Akdis CA, Arkwright PD, Brüggen MC, Busse W, Gadina M, Guttman-Yassky E, et al. (July 2020). "Type 2 immunity in the skin and lungs" (https://doi.org/10.1111%2Fall.14318). *Allergy*. **75** (7): 1582–1605. doi:10.1111/all.14318 (https://doi.org/10.1111%2Fall.14318). PMID 32319104 (https://pubmed.ncbi.nlm.nih.gov/32319104).
58. Eichenfield LF, Tom WL, Chamlin SL, Feldman SR, Hanifin JM, Simpson EL, et al. (February 2014). "Guidelines of care for the management of atopic dermatitis: section 1. Diagnosis and assessment of atopic dermatitis" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4410183). *Journal of the American Academy of Dermatology*. **70** (2): 338–351. doi:10.1016/j.jaad.2013.10.010 (https://doi.org/10.1016%2Fj.jaad.2013.10.010). PMC 4410183 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4410183). PMID 24290431 (https://pubmed.ncbi.nlm.nih.gov/24290431).
59. Brenninkmeijer EE, Schram ME, Leeflang MM, Bos JD, Spuls PI (April 2008). "Diagnostic criteria for atopic dermatitis: a systematic review" (https://doi.org/10.1111%2Fj.1365-2133.2007.08412.x). *The British Journal of Dermatology*. **158** (4): 754–765. doi:10.1111/j.1365-2133.2007.08412.x (https://doi.org/10.1111%2Fj.1365-2133.2007.08412.x). PMID 18241277 (https://pubmed.ncbi.nlm.nih.gov/18241277). S2CID 453564 (https://api.semanticscholar.org/CorpusID:453564).
60. Williams HC, Burney PG, Pembroke AC, Hay RJ (September 1994). "The U.K. Working Party's Diagnostic Criteria for Atopic Dermatitis. III. Independent hospital validation". *The British Journal of Dermatology*. **131** (3): 406–416. doi:10.1111/j.1365-2133.1994.tb08532.x (https://doi.org/10.1111%2Fj.1365-2133.1994.tb08532.x). PMID 7918017 (https://pubmed.ncbi.nlm.nih.gov/7918017). S2CID 37406163 (https://api.semanticscholar.org/CorpusID:37406163).
61. Trikamjee T, Comberiati P, D'Auria E, Peroni D, Zuccotti GV (12 January 2021). "Nutritional Factors in the Prevention of Atopic Dermatitis in Children" (https://doi.org/10.3389%2Ffped.2020.577413). *Frontiers in Pediatrics*. **8** 577413. doi:10.3389/fped.2020.577413 (https://doi.org/10.3389%2Ffped.2020.577413). PMC 7874114 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7874114). PMID 33585361 (https://pubmed.ncbi.nlm.nih.gov/33585361).
62. Sun S, Chang G, Zhang L (June 2022). "The prevention effect of probiotics against eczema in children: an update systematic review and meta-analysis". *The Journal of Dermatological Treatment*. **33** (4): 1844–1854. doi:10.1080/09546634.2021.1925077 (https://doi.org/10.1080%2F09546634.2021.1925077). PMID 34006167 (https://pubmed.ncbi.nlm.nih.gov/34006167).

63. Voigt J, Lele M (November 2022). "Lactobacillus rhamnosus Used in the Perinatal Period for the Prevention of Atopic Dermatitis in Infants: A Systematic Review and Meta-Analysis of Randomized Trials" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9576646). *American Journal of Clinical Dermatology*. **23** (6): 801–811. doi:10.1007/s40257-022-00723-x (https://doi.org/10.1007%2Fs40257-022-00723-x). PMC 9576646 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9576646). PMID 36161401 (https://pubmed.ncbi.nlm.nih.gov/36161401).
64. Kelleher MM, Phillips R, Brown SJ, Cro S, Cornelius V, Carlsen KC, et al. (Cochrane Skin Group) (November 2022). "Skin care interventions in infants for preventing eczema and food allergy" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9661877). *The Cochrane Database of Systematic Reviews*. **2022** (11) CD013534. doi:10.1002/14651858.CD013534.pub3 (https://doi.org/10.1002%2F14651858.CD013534.pub3). PMC 9661877 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9661877). PMID 36373988 (https://pubmed.ncbi.nlm.nih.gov/36373988).
65. "What are Topical Treatments for Eczema and How Should They Be Used?" (https://nationaleczema.org/eczema/treatment/topicals/). *National Eczema Association*. Retrieved 22 June 2023.
66. Chopra R, Silverberg JI (2018). "Assessing the severity of atopic dermatitis in clinical trials and practice". *Clinics in Dermatology*. Atopic Dermatits: Part II. **36** (5): 606–615. doi:10.1016/j.clindermatol.2018.05.012 (https://doi.org/10.1016%2Fj.clindermatol.2018.05.012). PMID 30217273 (https://pubmed.ncbi.nlm.nih.gov/30217273). S2CID 52277845 (https://api.semanticscholar.org/CorpusID:52277845).
67. Ridd MJ, Roberts A, Grindlay D, Williams HC (October 2019). "Which emollients are effective and acceptable for eczema in children?" (https://research-information.bris.ac.uk/en/publications/2c0e2127-710c-46a9-bd5c-89574b11c9ee). *BMJ*. **367** l5882. doi:10.1136/bmj.l5882 (https://doi.org/10.1136%2Fbmj.l5882). hdl:1983/2c0e2127-710c-46a9-bd5c-89574b11c9ee (https://hdl.handle.net/1983%2F2c0e2127-710c-46a9-bd5c-89574b11c9ee). PMID 31649114 (https://pubmed.ncbi.nlm.nih.gov/31649114). S2CID 204882682 (https://api.semanticscholar.org/CorpusID:204882682).
68. Ridd MJ, Santer M, MacNeill SJ, Sanderson E, Wells S, Webb D, et al. (August 2022). "Effectiveness and safety of lotion, cream, gel, and ointment emollients for childhood eczema: a pragmatic, randomised, phase 4, superiority trial" (https://doi.org/10.1016%2FS2352-4642%2822%2900146-8). *The Lancet. Child & Adolescent Health*. **6** (8): 522–532. doi:10.1016/S2352-4642(22)00146-8 (https://doi.org/10.1016%2FS2352-4642%2822%2900146-8). hdl:1983/e4009d3c-127f-4aa9-bf71-e40401b33eee (https://hdl.handle.net/1983%2Fe4009d3c-127f-4aa9-bf71-e40401b33eee). PMID 35617974 (https://pubmed.ncbi.nlm.nih.gov/35617974). S2CID 249024141 (https://api.semanticscholar.org/CorpusID:249024141).
69. Chu DK, Chu AW, Rayner DG, Guyatt GH, Yepes-Nuñez JJ, Gomez-Escobar L, et al. (December 2023). "Topical treatments for atopic dermatitis (eczema): Systematic review and network meta-analysis of randomized trials". *The Journal of Allergy and Clinical Immunology*. **152** (6): 1493–1519. doi:10.1016/j.jaci.2023.08.030 (https://doi.org/10.1016%2Fj.jaci.2023.08.030). hdl:10576/50632 (https://hdl.handle.net/10576%2F50632). PMID 37678572 (https://pubmed.ncbi.nlm.nih.gov/37678572). S2CID 261610152 (https://api.semanticscholar.org/CorpusID:261610152).
70. *Evidence reviews for adding bath emollients to the management of atopic eczema in children under 12 years: Atopic eczema in under 12s: diagnosis and management: Evidence review A* (https://www.ncbi.nlm.nih.gov/books/NBK594326/). NICE Evidence Reviews Collection. London: National Institute for Health and Care Excellence (NICE). 2023. ISBN 978-1-4731-5235-9. PMID 37616434 (https://pubmed.ncbi.nlm.nih.gov/37616434).

71. Santer M, Ridd MJ, Francis NA, Stuart B, Rumsby K, Chorozoglou M, et al. (May 2018). "Emollient bath additives for the treatment of childhood eczema (BATHE): multicentre pragmatic parallel group randomised controlled trial of clinical and cost effectiveness" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5930266). *BMJ*. **361** k1332. doi:10.1136/bmj.k1332 (https://doi.org/10.1136%2Fbmj.k1332). PMC 5930266 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5930266). PMID 29724749 (https://pubmed.ncbi.nlm.nih.gov/29724749).
72. Lax SJ, Harvey J, Axon E, Howells L, Santer M, Ridd MJ, et al. (Cochrane Skin Group) (March 2022). "Strategies for using topical corticosteroids in children and adults with eczema" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8916090). *The Cochrane Database of Systematic Reviews*. **2022** (3) CD013356. doi:10.1002/14651858.CD013356.pub2 (https://doi.org/10.1002%2F14651858.CD013356.pub2). PMC 8916090 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8916090). PMID 35275399 (https://pubmed.ncbi.nlm.nih.gov/35275399).
73. Harvey J, Lax SJ, Lowe A, Santer M, Lawton S, Langan SM, et al. (October 2023). "The long-term safety of topical corticosteroids in atopic dermatitis: A systematic review" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10549798). *Skin Health and Disease*. **3** (5) ski2.268: e268. doi:10.1002/ski2.268 (https://doi.org/10.1002%2Fski2.268). PMC 10549798 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10549798). PMID 37799373 (https://pubmed.ncbi.nlm.nih.gov/37799373).
74. Axon E, Chalmers JR, Santer M, Ridd MJ, Lawton S, Langan SM, et al. (July 2021). "Safety of topical corticosteroids in atopic eczema: an umbrella review" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8264889). *BMJ Open*. **11** (7) e046476. doi:10.1136/bmjopen-2020-046476 (https://doi.org/10.1136%2Fbmjopen-2020-046476). PMC 8264889 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8264889). PMID 34233978 (https://pubmed.ncbi.nlm.nih.gov/34233978).
75. Chu DK, Schneider L, Asiniwasis RN, Boguniewicz M, De Benedetto A, Ellison K, et al. (March 2024). "Atopic dermatitis (eczema) guidelines: 2023 American Academy of Allergy, Asthma and Immunology/American College of Allergy, Asthma and Immunology Joint Task Force on Practice Parameters GRADE- and Institute of Medicine-based recommendations" (https://doi.org/10.1016%2Fj.anai.2023.11.009). *Annals of Allergy, Asthma & Immunology*. **132** (3): 274–312. doi:10.1016/j.anai.2023.11.009 (https://doi.org/10.1016%2Fj.anai.2023.11.009). PMID 38108679 (https://pubmed.ncbi.nlm.nih.gov/38108679).
76. Cury Martins J, Martins C, Aoki V, Gois AF, Ishii HA, da Silva EM (July 2015). "Topical tacrolimus for atopic dermatitis" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6461158). *The Cochrane Database of Systematic Reviews*. **2015** (7) CD009864. doi:10.1002/14651858.CD009864.pub2 (https://doi.org/10.1002%2F14651858.CD009864.pub2). PMC 6461158 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6461158). PMID 26132597 (https://pubmed.ncbi.nlm.nih.gov/26132597).
77. Devasenapathy N, Chu A, Wong M, Srivastava A, Ceccacci R, Lin C, et al. (January 2023). "Cancer risk with topical calcineurin inhibitors, pimecrolimus and tacrolimus, for atopic dermatitis: a systematic review and meta-analysis". *The Lancet. Child & Adolescent Health*. **7** (1): 13–25. doi:10.1016/S2352-4642(22)00283-8 (https://doi.org/10.1016%2FS2352-4642%2822%2900283-8). PMID 36370744 (https://pubmed.ncbi.nlm.nih.gov/36370744). S2CID 253470127 (https://api.semanticscholar.org/CorpusID:253470127).
78. McDowell L, Olin B (August 2019). "Crisaborole: A Novel Nonsteroidal Topical Treatment for Atopic Dermatitis" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6600556). *The Journal of Pharmacy Technology*. **35** (4): 172–178. doi:10.1177/8755122519844507 (https://doi.org/10.1177%2F8755122519844507). PMC 6600556 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6600556). PMID 34861031 (https://pubmed.ncbi.nlm.nih.gov/34861031).

79. He Y, Liu J, Wang Y, Kuai W, Liu R, Wu J (6 February 2023). Pimpinelli N (ed.). "Topical Administration of Crisaborole in Mild to Moderate Atopic Dermatitis: A Systematic Review and Meta-Analysis" (https://doi.org/10.1155%2F2023%2F1869934). *Dermatologic Therapy*. **2023** 1869934: 1–9. doi:10.1155/2023/1869934 (https://doi.org/10.1155%2F2023%2F1869934). ISSN 1529-8019 (https://search.worldcat.org/issn/1529-8019).
80. "Otsuka's Moizerto Ointment Granted Approval in Japan as a Treatment for Atopic Dermatitis" (https://www.otsuka.co.jp/en/company/newsreleases/2021/20210927_1.html). *Otsuka Pharmaceutical* (Press release). 27 September 2021. Retrieved 15 February 2026.
81. "Acrotech Biopharma Inc., Announces FDA Approval of Adquey (difamilast 1%) Ointment for the Treatment of Mild-to-Moderate Atopic Dermatitis" (https://www.aurobindo.com/api/uploads/disclosureofeventsinformation/LtrToSEsDifamilastNDAapprovalAcrotech13022026.pdf) (PDF) (Press release). 13 February 2026. Retrieved 15 February 2026.
82. Davis DM, Drucker AM, Alikhan A, Bercovitch L, Cohen DE, Darr JM, et al. (February 2024). "Guidelines of care for the management of atopic dermatitis in adults with phototherapy and systemic therapies". *Journal of the American Academy of Dermatology*. **90** (2): e43–e56. doi:10.1016/j.jaad.2023.08.102 (https://doi.org/10.1016%2Fj.jaad.2023.08.102). PMID 37943240 (https://pubmed.ncbi.nlm.nih.gov/37943240).
83. Paolino A, Alexander H, Broderick C, Flohr C (May 2023). "Non-biologic systemic treatments for atopic dermatitis: Current state of the art and future directions" (https://doi.org/10.1111%2Fcea.14301). *Clinical and Experimental Allergy*. **53** (5): 495–510. doi:10.1111/cea.14301 (https://doi.org/10.1111%2Fcea.14301). PMID 36949024 (https://pubmed.ncbi.nlm.nih.gov/36949024).
84. Flohr C, Rosala-Hallas A, Jones AP, Beattie P, Baron S, Browne F, et al. (November 2023). "Efficacy and safety of ciclosporin versus methotrexate in the treatment of severe atopic dermatitis in children and young people (TREAT): a multicentre parallel group assessor-blinded clinical trial". *The British Journal of Dermatology*. **189** (6): 674–684. doi:10.1093/bjd/ljad281 (https://doi.org/10.1093%2Fbjd%2Fljad281). PMID 37722926 (https://pubmed.ncbi.nlm.nih.gov/37722926).
85. Kim K (November 2012). "Neuroimmunological mechanism of pruritus in atopic dermatitis focused on the role of serotonin" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3762292). *Biomolecules & Therapeutics*. **20** (6): 506–512. doi:10.4062/biomolther.2012.20.6.506 (https://doi.org/10.4062%2Fbiomolther.2012.20.6.506). PMC 3762292 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3762292). PMID 24009842 (https://pubmed.ncbi.nlm.nih.gov/24009842).
86. Chu AW, Wong MM, Rayner DG, Guyatt GH, Díaz Martinez JP, Ceccacci R, et al. (December 2023). "Systemic treatments for atopic dermatitis (eczema): Systematic review and network meta-analysis of randomized trials" (https://doi.org/10.1016%2Fj.jaci.2023.08.029). *The Journal of Allergy and Clinical Immunology*. **152** (6): 1470–1492. doi:10.1016/j.jaci.2023.08.029 (https://doi.org/10.1016%2Fj.jaci.2023.08.029). PMID 37678577 (https://pubmed.ncbi.nlm.nih.gov/37678577).
87. "FDA approves new eczema drug Dupixent" (https://web.archive.org/web/20170328204026/https://www.fda.gov/NewsEvents/Newsroom/PressAnnouncements/ucm549078.htm). US Food & Drug Administration. 28 March 2017. Archived from the original (https://www.fda.gov/NewsEvents/Newsroom/PressAnnouncements/ucm549078.htm) on 28 March 2017. Retrieved 29 March 2017.
88. "Dupixent" (https://www.ema.europa.eu/en/medicines/human/EPAR/dupixent). *European Medicines Agency*. 17 September 2018. Retrieved 22 March 2023.
89. "Adtralza EPAR" (https://www.ema.europa.eu/en/medicines/human/EPAR/adtralza). *European Medicines Agency (EMA)*. 20 April 2021. Retrieved 9 July 2021. Text was copied from this source which is copyright European Medicines Agency. Reproduction is authorized provided the source is acknowledged.

90. "Drug Approval Package: ADBRY" (https://web.archive.org/web/20220803033420/https://www.accessdata.fda.gov/drugsatfda_docs/nda/2022/761180Orig1s000TOC.cfm). US Food & Drug Administration. 27 December 2021. Archived from the original (https://www.accessdata.fda.gov/drugsatfda_docs/nda/2022/761180Orig1s000TOC.cfm) on 3 August 2022. Retrieved 6 March 2022.
91. "Ebglyss" (https://www.ema.europa.eu/en/medicines/human/EPAR/ebglyss). *European Medicines Agency*. 21 November 2023. Retrieved 15 April 2024.
92. "FDA Rejects Lilly's Eczema Treatment Over Third-Party Manufacturing Issues" (https://www.biospace.com/article/fda-rejects-lilly-s-eczema-treatment-over-third-party-manufacturing-issues/). *BioSpace*. 2 October 2023. Retrieved 3 October 2023.
93. "U.S. FDA Approves Pfizer's Cibinqo (abrocitinib) for Adults with Moderate-to-Severe Atopic Dermatitis" (https://www.pfizer.com/news/press-release/press-release-detail/us-fda-approves-pfizers-cibinqor-abrocitinib-adults). *Pfizer Inc.* (Press release). 14 January 2022. Retrieved 16 January 2022.
94. "U.S. FDA Approves Rinvoq (upadacitinib) to Treat Adults and Children 12 Years and Older with Refractory, Moderate to Severe Atopic Dermatitis" (https://news.abbvie.com/news/press-releases/us-fda-approves-rinvoq-upadacitinib-to-treat-adults-and-children-12-years-and-older-with-refractory-moderate-to-severe-atopic-dermatitis.htm). *AbbeVie* (Press release). Retrieved 6 March 2022.
95. "NEMLUVIO® (nemolizumab-ilto) for injection, for subcutaneous use Initial U.S. Approval: 2024" (https://www.accessdata.fda.gov/drugsatfda_docs/label/2024/761391s000lbl.pdf) (PDF). *www.accessdata.fda.gov*.
96. Yepes-Nuñez JJ, Guyatt GH, Gómez-Escobar LG, Pérez-Herrera LC, Chu AW, Ceccaci R, et al. (January 2023). "Allergen immunotherapy for atopic dermatitis: Systematic review and meta-analysis of benefits and harms" (https://doi.org/10.1016%2Fj.jaci.2022.09.020). *The Journal of Allergy and Clinical Immunology*. **151** (1): 147–158. doi:10.1016/j.jaci.2022.09.020 (https://doi.org/10.1016%2Fj.jaci.2022.09.020). hdl:10576/44628 (https://hdl.handle.net/10576%2F44628). PMID 36191689 (https://pubmed.ncbi.nlm.nih.gov/36191689). S2CID 252656283 (https://api.semanticscholar.org/CorpusID:252656283).
97. Tam H, Calderon MA, Manikam L, Nankervis H, García Núñez I, Williams HC, et al. (February 2016). "Specific allergen immunotherapy for the treatment of atopic eczema" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8761476). *The Cochrane Database of Systematic Reviews*. **2016** (2) CD008774. doi:10.1002/14651858.CD008774.pub2 (https://doi.org/10.1002%2F14651858.CD008774.pub2). hdl:10044/1/31818 (https://hdl.handle.net/10044%2F1%2F31818). PMC 8761476 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8761476). PMID 26871981 (https://pubmed.ncbi.nlm.nih.gov/26871981).
98. "Secondary bacterial infection of eczema and other common skin conditions: antimicrobial prescribing. NICE guideline [NG190]" (https://www.nice.org.uk/guidance/ng190/chapter/Recommendations). *National Institute for Health and Care Excellence*. 2 March 2021. Retrieved 26 July 2024.
99. Dębińska A, Sikorska-Szaflik H, Urbanik M, Boznański A (2015). "The role of vitamin D in atopic dermatitis". *Dermatitis* (Review). **26** (4): 155–161. doi:10.1097/DER.0000000000000128 (https://doi.org/10.1097%2FDER.0000000000000128). PMID 26172483 (https://pubmed.ncbi.nlm.nih.gov/26172483). S2CID 35345939 (https://api.semanticscholar.org/CorpusID:35345939).
100. Kim G, Bae JH (September 2016). "Vitamin D and atopic dermatitis: A systematic review and meta-analysis". *Nutrition* (Systematic Review and Meta-Analysis). **32** (9): 913–920. doi:10.1016/j.nut.2016.01.023 (https://doi.org/10.1016%2Fj.nut.2016.01.023). PMID 27061361 (https://pubmed.ncbi.nlm.nih.gov/27061361).

101. Hattangdi-Haridas SR, Lanham-New SA, Wong WH, Ho MH, Darling AL (August 2019). "Vitamin D Deficiency and Effects of Vitamin D Supplementation on Disease Severity in Patients with Atopic Dermatitis: A Systematic Review and Meta-Analysis in Adults and Children" (https://doi.org/10.3390%2Fnu11081854). *Nutrients*. **11** (8): 1854. doi:10.3390/nu11081854 (https://doi.org/10.3390%2Fnu11081854). PMC 6722944 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6722944). PMID 31405041 (https://pubmed.ncbi.nlm.nih.gov/31405041).
102. Venter C, Agostoni C, Arshad SH, Ben-Abdallah M, Du Toit G, Fleischer DM, et al. (November 2020). Peters R (ed.). "Dietary factors during pregnancy and atopic outcomes in childhood: A systematic review from the European Academy of Allergy and Clinical Immunology" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9588404). *Pediatric Allergy and Immunology*. **31** (8): 889–912. doi:10.1111/pai.13303 (https://doi.org/10.1111%2Fpai.13303). PMC 9588404 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9588404). PMID 32524677 (https://pubmed.ncbi.nlm.nih.gov/32524677).
103. Trikamjee T, Comberiati P, D'Auria E, Peroni D, Zuccotti GV (12 January 2021). "Nutritional Factors in the Prevention of Atopic Dermatitis in Children" (https://doi.org/10.3389%2Ffped.2020.577413). *Frontiers in Pediatrics*. **8** 577413. doi:10.3389/fped.2020.577413 (https://doi.org/10.3389%2Ffped.2020.577413). PMC 7874114 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7874114). PMID 33585361 (https://pubmed.ncbi.nlm.nih.gov/33585361).
104. Makrgeorgou A, Leonardi-Bee J, Bath-Hextall FJ, Murrell DF, Tang ML, Roberts A, et al. (November 2018). "Probiotics for treating eczema" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6517242). *The Cochrane Database of Systematic Reviews*. **2018** (11) CD006135. doi:10.1002/14651858.CD006135.pub3 (https://doi.org/10.1002%2F14651858.CD006135.pub3). PMC 6517242 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6517242). PMID 30480774 (https://pubmed.ncbi.nlm.nih.gov/30480774).
105. Rusu E, Enache G, Cursaru R, Alexescu A, Radu R, Onila O, et al. (August 2019). "Prebiotics and probiotics in atopic dermatitis" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6639913). *Experimental and Therapeutic Medicine*. **18** (2): 926–931. doi:10.3892/etm.2019.7678 (https://doi.org/10.3892%2Fetm.2019.7678). PMC 6639913 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6639913). PMID 31384325 (https://pubmed.ncbi.nlm.nih.gov/31384325).
106. Umborowati MA, Damayanti D, Anggraeni S, Endaryanto A, Surono IS, Effendy I, et al. (August 2022). "The role of probiotics in the treatment of adult atopic dermatitis: a meta-analysis of randomized controlled trials" (https://doi.org/10.1186%2Fs41043-022-00318-6). *Journal of Health, Population, and Nutrition*. **41** (1) 37. Bibcode:2022JHPN...41...37U (https://ui.adsabs.harvard.edu/abs/2022JHPN...41...37U). doi:10.1186/s41043-022-00318-6 (https://doi.org/10.1186%2Fs41043-022-00318-6). PMC 9386980 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9386980). PMID 35978397 (https://pubmed.ncbi.nlm.nih.gov/35978397).
107. Chang YS, Trivedi MK, Jha A, Lin YF, Dimaano L, García-Romero MT (March 2016). "Synbiotics for Prevention and Treatment of Atopic Dermatitis: A Meta-analysis of Randomized Clinical Trials" (https://doi.org/10.1001%2Fjamapediatrics.2015.3943). *JAMA Pediatrics*. **170** (3): 236–242. doi:10.1001/jamapediatrics.2015.3943 (https://doi.org/10.1001%2Fjamapediatrics.2015.3943). PMID 26810481 (https://pubmed.ncbi.nlm.nih.gov/26810481).
108. Roberts K, Gilbertson A, Dawson S, Turner N, Ridd MJ (March 2022). "Test-guided dietary exclusions for treating established atopic dermatitis in children: A systematic review" (https://research-information.bris.ac.uk/en/publications/693f4124-283d-4a11-adb3-21e1916330be). *Clinical and Experimental Allergy*. **52** (3): 442–446. doi:10.1111/cea.14072 (https://doi.org/10.1111%2Fcea.14072). hdl:1983/693f4124-283d-4a11-adb3-21e1916330be (https://hdl.handle.net/1983%2F693f4124-283d-4a11-adb3-21e1916330be). PMID 34862822 (https://pubmed.ncbi.nlm.nih.gov/34862822). S2CID 244872550 (https://api.semanticscholar.org/CorpusID:244872550).

109. Lio PA (October 2013). "Non-pharmacologic therapies for atopic dermatitis". *Current Allergy and Asthma Reports*. **13** (5): 528–538. doi:10.1007/s11882-013-0371-y (https://doi.org/10.1007%2Fs11882-013-0371-y). PMID 23881511 (https://pubmed.ncbi.nlm.nih.gov/23881511). S2CID 40875822 (https://api.semanticscholar.org/CorpusID:40875822).
110. Bakaa L, Pernica JM, Couban RJ, Tackett KJ, Burkhart CN, Leins L, et al. (June 2022). "Bleach baths for atopic dermatitis: A systematic review and meta-analysis including unpublished data, Bayesian interpretation, and GRADE" (https://doi.org/10.1016%2Fj.anai.2022.03.024). *Annals of Allergy, Asthma & Immunology*. **128** (6): 660–668.e9. doi:10.1016/j.anai.2022.03.024 (https://doi.org/10.1016%2Fj.anai.2022.03.024). PMID 35367346 (https://pubmed.ncbi.nlm.nih.gov/35367346). S2CID 247847598 (https://api.semanticscholar.org/CorpusID:247847598).
111. Jaros J, Wilson C, Shi VY (August 2020). "Fabric Selection in Atopic Dermatitis: An Evidence-Based Review". *American Journal of Clinical Dermatology*. **21** (4): 467–482. doi:10.1007/s40257-020-00516-0 (https://doi.org/10.1007%2Fs40257-020-00516-0). PMID 32440827 (https://pubmed.ncbi.nlm.nih.gov/32440827). S2CID 218761019 (https://api.semanticscholar.org/CorpusID:218761019).
112. "Clothing and eczema" (https://eczema.org/information-and-advice/triggers-for-eczema/clothing-and-eczema/). *National Eczema Society*. 11 February 2020. Retrieved 10 April 2023.
113. Thomas KS, Bradshaw LE, Sach TH, Cowdell F, Batchelor JM, Lawton S, et al. (April 2017). "Randomised controlled trial of silk therapeutic garments for the management of atopic eczema in children: the CLOTHES trial" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5410632). *Health Technology Assessment*. **21** (16): 1–260. doi:10.3310/hta21160 (https://doi.org/10.3310%2Fhta21160). PMC 5410632 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5410632). PMID 28409557 (https://pubmed.ncbi.nlm.nih.gov/28409557).
114. Ridd MJ, King AJ, Le Roux E, Waldecker A, Huntley AL (September 2017). "Systematic review of self-management interventions for people with eczema" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5637890). *The British Journal of Dermatology*. **177** (3): 719–734. doi:10.1111/bjd.15601 (https://doi.org/10.1111%2Fbjd.15601). PMC 5637890 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5637890). PMID 28432696 (https://pubmed.ncbi.nlm.nih.gov/28432696).
115. Teasdale E, Muller I, Sivyer K, Ghio D, Greenwell K, Wilczynska S, et al. (April 2021). "Views and experiences of managing eczema: systematic review and thematic synthesis of qualitative studies". *The British Journal of Dermatology*. **184** (4): 627–637. doi:10.1111/bjd.19299 (https://doi.org/10.1111%2Fbjd.19299). hdl:1983/05ffa4c9-3868-469d-97b6-ac467bdf95b3 (https://hdl.handle.net/1983%2F05ffa4c9-3868-469d-97b6-ac467bdf95b3). PMID 32531800 (https://pubmed.ncbi.nlm.nih.gov/32531800).
116. Addressing misconceptions about eczema could help people manage their condition over the long term (https://evidence.nihr.ac.uk/alert/eczema-misconceptions-hinder-long-term-treatment/) (Report). NIHR Evidence. 9 December 2020. doi:10.3310/alert_42973 (https://doi.org/10.3310%2Falert_42973).
117. Thandi CS, Constantinou S, Vincent R, Ridd MJ (June 2023). "Where and how have written action plans for atopic eczema/dermatitis been developed and evaluated? Systematic review" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10233085). *Skin Health and Disease*. **3** (3) ski2.213: e213. doi:10.1002/ski2.213 (https://doi.org/10.1002%2Fski2.213). PMC 10233085 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10233085). PMID 37275422 (https://pubmed.ncbi.nlm.nih.gov/37275422). S2CID 257729897 (https://api.semanticscholar.org/CorpusID:257729897).
118. "Online support improved eczema symptoms in children and young people" (https://evidence.nihr.ac.uk/alert/online-support-improved-eczema-symptoms-in-children-and-young-people/). *NIHR Evidence* (Plain English summary). National Institute for Health and Care Research. 11 April 2023. doi:10.3310/nihrevidence_57579 (https://doi.org/10.3310%2Fnihrevidence_57579). S2CID 258094184 (https://api.semanticscholar.org/CorpusID:258094184).

119. Santer M, Muller I, Becque T, Stuart B, Hooper J, Steele M, et al. (December 2022). "Eczema Care Online behavioural interventions to support self-care for children and young people: two independent, pragmatic, randomised controlled trials" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11778922). *BMJ*. **379** e072007. doi:10.1136/bmj-2022-072007 (https://doi.org/10.1136%2Fbmj-2022-072007). hdl:1983/6003736c-fa64-47a6-9ca7-984ba6a3bad1 (https://hdl.handle.net/1983%2F6003736c-fa64-47a6-9ca7-984ba6a3bad1). PMC 11778922 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11778922). PMID 36740888 (https://pubmed.ncbi.nlm.nih.gov/36740888). S2CID 254367009 (https://api.semanticscholar.org/CorpusID:254367009).
120. Tintle S, Shemer A, Suárez-Fariñas M, Fujita H, Gilleaudeau P, Sullivan-Whalen M, et al. (September 2011). "Reversal of atopic dermatitis with narrow-band UVB phototherapy and biomarkers for therapeutic response" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3448950). *The Journal of Allergy and Clinical Immunology*. **128** (3): 583–93.e1–4. doi:10.1016/j.jaci.2011.05.042 (https://doi.org/10.1016%2Fj.jaci.2011.05.042). PMC 3448950 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3448950). PMID 21762976 (https://pubmed.ncbi.nlm.nih.gov/21762976).
121. Beattie PE, Finlan LE, Kernohan NM, Thomson G, Hupp TR, Ibbotson SH (May 2005). "The effect of ultraviolet (UV) A1, UVB and solar-simulated radiation on p53 activation and p21". *The British Journal of Dermatology*. **152** (5): 1001–1008. doi:10.1111/j.1365-2133.2005.06557.x (https://doi.org/10.1111%2Fj.1365-2133.2005.06557.x). PMID 15888160 (https://pubmed.ncbi.nlm.nih.gov/15888160). S2CID 22191753 (https://api.semanticscholar.org/CorpusID:22191753).
122. Musters AH, Mashayekhi S, Harvey J, Axon E, Lax SJ, Flohr C, et al. (Cochrane Skin Group) (October 2021). "Phototherapy for atopic eczema" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8552896). *The Cochrane Database of Systematic Reviews*. **10** (10) CD013870. doi:10.1002/14651858.CD013870.pub2 (https://doi.org/10.1002%2F14651858.CD013870.pub2). PMC 8552896 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8552896). PMID 34709669 (https://pubmed.ncbi.nlm.nih.gov/34709669).
123. Jans J, Garinis GA, Schul W, van Oudenaren A, Moorhouse M, Smid M, et al. (November 2006). "Differential role of basal keratinocytes in UV-induced immunosuppression and skin cancer" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1636796). *Molecular and Cellular Biology*. **26** (22): 8515–8526. doi:10.1128/MCB.00807-06 (https://doi.org/10.1128%2FMCB.00807-06). PMC 1636796 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1636796). PMID 16966369 (https://pubmed.ncbi.nlm.nih.gov/16966369).
124. Gu S, Yang AW, Xue CC, Li CG, Pang C, Zhang W, et al. (September 2013). "Chinese herbal medicine for atopic eczema" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10639001). *The Cochrane Database of Systematic Reviews*. **2013** (9) CD008642. doi:10.1002/14651858.CD008642.pub2 (https://doi.org/10.1002%2F14651858.CD008642.pub2). PMC 10639001 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10639001). PMID 24018636 (https://pubmed.ncbi.nlm.nih.gov/24018636).
125. Park J, Lio P (18 March 2025). "Patient Reported Outcome Measures in Complementary & Alternative Medicine for Atopic Dermatitis: A Systematic Review" (https://jintegrativederm.org/index.php/joid/article/view/46). *Journal of Integrative Dermatology*. doi:10.64550/joid.xpgfjg26 (https://doi.org/10.64550%2Fjoid.xpgfjg26) (inactive 14 January 2026).
126. Fasseeh AN, Elezbawy B, Korra N, Tannira M, Dalle H, Aderian S, et al. (December 2022). "Burden of Atopic Dermatitis in Adults and Adolescents: a Systematic Literature Review" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9674816). *Dermatology and Therapy*. **12** (12): 2653–2668. doi:10.1007/s13555-022-00819-6 (https://doi.org/10.1007%2Fs13555-022-00819-6). PMC 9674816 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9674816). PMID 36197589 (https://pubmed.ncbi.nlm.nih.gov/36197589).

127. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. (March 2021). "The PRISMA 2020 statement: an updated guideline for reporting systematic reviews" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8005924). *BMJ*. **372**: n71. doi:10.1136/bmj.n71 (https://doi.org/10.1136%2Fbmj.n71). PMC 8005924 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8005924). PMID 33782057 (https://pubmed.ncbi.nlm.nih.gov/33782057).
128. Marron SE, Cebrian-Rodriguez J, Alcalde-Herrero VM, Garcia-Latasa de Aranibar FJ, Tomas-Aragones L (1 July 2020). "Psychosocial Impact of Atopic Dermatitis in Adults: A Qualitative Study" (https://doi.org/10.1016%2Fj.ad.2019.03.018). *Actas Dermo-Sifiliograficas*. **111** (6): 513–517. doi:10.1016/j.ad.2019.03.018 (https://doi.org/10.1016%2Fj.ad.2019.03.018). PMID 32401725 (https://pubmed.ncbi.nlm.nih.gov/32401725).
129. Silverwood RJ, Mansfield KE, Mulick A, Wong AY, Schmidt SA, Roberts A, et al. (May 2021). "Atopic eczema in adulthood and mortality: UK population-based cohort study, 1998-2016" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8098860). *The Journal of Allergy and Clinical Immunology*. **147** (5): 1753–1763. doi:10.1016/j.jaci.2020.12.001 (https://doi.org/10.1016%2Fj.jaci.2020.12.001). PMC 8098860 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8098860). PMID 33516523 (https://pubmed.ncbi.nlm.nih.gov/33516523).
130. Pedersen CJ, Uddin MJ, Saha SK, Darmstadt GL (16 April 2021). "Prevalence and psychosocial impact of atopic dermatitis in Bangladeshi children and families" (https://doi.org/10.1371%2Fjournal.pone.0249824). *PLOS ONE*. **16** (4) e0249824. Bibcode:2021PLoSO..1649824P (https://ui.adsabs.harvard.edu/abs/2021PLoSO..1649824P). doi:10.1371/journal.pone.0249824 (https://doi.org/10.1371%2Fjournal.pone.0249824). PMC 8051797 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8051797). PMID 33861780 (https://pubmed.ncbi.nlm.nih.gov/33861780).
131. "World Bank Country and Lending Groups – World Bank Data Help Desk" (https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups). *datahelpdesk.worldbank.org*. Retrieved 14 January 2024.
132. Avena-Woods C (June 2017). "Overview of atopic dermatitis" (https://pubmed.ncbi.nlm.nih.gov/28978208/?dopt=Abstract). *The American Journal of Managed Care*. **23** (8 Suppl): S115–S123. PMID 28978208 (https://pubmed.ncbi.nlm.nih.gov/28978208).
133. Elezbawy B, Fasseeh AN, Fouly E, Tannira M, Dalle H, Aderian S, et al. (January 2023). "Humanistic and Economic Burden of Atopic Dermatitis for Adults and Adolescents in the Middle East and Africa Region" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9823172). *Dermatology and Therapy*. **13** (1): 131–146. doi:10.1007/s13555-022-00857-0 (https://doi.org/10.1007%2Fs13555-022-00857-0). PMC 9823172 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9823172). PMID 36445612 (https://pubmed.ncbi.nlm.nih.gov/36445612).
134. Elezbawy B, Farghaly M, Al Lafi A, Gamal M, Metni M, Visser W, et al. (July 2024). "Strategic Approaches to Reducing the Burden of Atopic Dermatitis in the Middle East and Africa Region" (https://doi.org/10.1016%2Fj.vhri.2024.100987). *Value in Health Regional Issues*. **42** 100987. doi:10.1016/j.vhri.2024.100987 (https://doi.org/10.1016%2Fj.vhri.2024.100987). PMID 38703753 (https://pubmed.ncbi.nlm.nih.gov/38703753).
135. Saito H (August 2005). "Much atopy about the skin: genome-wide molecular analysis of atopic eczema". *International Archives of Allergy and Immunology*. **137** (4): 319–325. doi:10.1159/000086464 (https://doi.org/10.1159%2F000086464). PMID 15970641 (https://pubmed.ncbi.nlm.nih.gov/15970641). S2CID 20040720 (https://api.semanticscholar.org/CorpusID:20040720).
136. "Atopic Dermatitis" (https://web.archive.org/web/20150408150538/http://www.uchospitals.edu/online-library/content%3DP01675). *www.uchospitals.edu*. 1 January 2015. Archived from the original (http://www.uchospitals.edu/online-library/content=P01675) on 8 April 2015. Retrieved 2 April 2015.

137. O'Connor C, Murphy M (December 2021). "Scratching the surface: a review of online misinformation and conspiracy theories in atopic dermatitis". *Clinical and Experimental Dermatology*. **46** (8): 1545–1547. doi:10.1111/ced.14679 (https://doi.org/10.1111%2Fced.14679). hdl:10468/11402 (https://hdl.handle.net/10468%2F11402). PMID 33864398 (https://pubmed.ncbi.nlm.nih.gov/33864398). S2CID 233278383 (https://api.semanticscholar.org/CorpusID:233278383).
138. Chin WK, Lee SW (October 2018). "A systematic review on the off-label use of montelukast in atopic dermatitis treatment". *International Journal of Clinical Pharmacy*. **40** (5): 963–976. doi:10.1007/s11096-018-0655-3 (https://doi.org/10.1007%2Fs11096-018-0655-3). PMID 29777328 (https://pubmed.ncbi.nlm.nih.gov/29777328). S2CID 21753181 (https://api.semanticscholar.org/CorpusID:21753181).
139. Ferguson L, Futamura M, Vakirlis E, Kojima R, Sasaki H, Roberts A, et al. (October 2018). "Leukotriene receptor antagonists for eczema" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6517006). *The Cochrane Database of Systematic Reviews*. **2018** (10) CD011224. doi:10.1002/14651858.cd011224.pub2 (https://doi.org/10.1002%2F14651858.cd011224.pub2). PMC 6517006 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6517006). PMID 30343498 (https://pubmed.ncbi.nlm.nih.gov/30343498).
140. Toledo LM, Rodriguez R, Sivesind TE, Vakirlis E, Kojima R, Dellavalle RP (April 2024). "From the Cochrane Library: Leukotriene Receptor Antagonists for Eczema" (https://doi.org/10.2196%2F50434). *JMIR Dermatology*. **7** e50434. doi:10.2196/50434 (https://doi.org/10.2196%2F50434). PMC 11053388 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11053388). PMID 38607671 (https://pubmed.ncbi.nlm.nih.gov/38607671).

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